

GRIFFIN SHEA

(613)-276-8063 | Ottawa, ON

griffin.shea@danjo.ca | <https://www.linkedin.com/in/griffinshea2022/>

SUMMARY

Reliable and motivated professional with a Bachelor of Computer Science and two years of experience in supporting mission-critical mainframe systems for the Canadian government. Focused on identifying and analyzing complex problems as well as designing and implementing robust programming solutions. Confident in producing quality work independently or as part of a committed team. Employs a positive attitude on the job and takes pride in a job well done. Holds Secret (level II) security clearance with the Canadian government.

SKILLS & ABILITIES

Technical

- Proficient programmer using COBOL, Python, C, C#, C++, Java, and JavaScript
- Two years of experience supporting mission-critical Unisys MCP mainframe systems for the Canadian government working with COBOL, WFL, and DMSII
- Earned IBM Z Xplore badges: Concepts, Fundamentals

Analytical

- Performed algorithm analysis to identify weak points in code, design optimizations, and compare solutions
- Studied and compared machine learning algorithms on time and space complexity, bias and variance tradeoffs, learning curves, and other metrics to choose optimal solutions to given problems
- Solid understanding of object-oriented software engineering principles and design patterns and their purpose and proper application

Communication

- Researched advanced topics in computer science and produced development projects and technical reports to share relevant findings with peers
 - Familiar with Agile software development methodologies and scrum framework
-

WORK EXPERIENCE

Designer

October 2024 – Present

Electors Team, Ottawa ON

- Designing a card game through rapid prototyping, applying game balance analysis and multi-agent game theory to make weekly improvements to a novel tactical negotiation game for four players
- Organizing supervised playtesting sessions, accounting for a diversity of personal needs and scheduling requirements to regularly assemble multiple participants on a weekly basis
- Collecting valuable insights from end-users through thoughtful and open-ended questioning as well as observing product use to identify problem areas and plan improvements for the next iteration

System Analyst 3

February 2024 – October 2024

Unisys Canada, Ottawa ON

- Worked closely with colleagues in a team-oriented work environment by participating in several meetings each week to establish effective working relationships and ensure goals are met in a timely manner
- Provided third-level support for mission-critical MCP systems, adhering to corporate policies, practices, and procedures to ensure high standards of quality and accuracy in delivering service to clients
- Tracked technical investigations through ticketing software, issuing prompt responses in clear language and collaborating with colleagues between departments to investigate problems and provide solutions for clients

- Implemented upgrades by creating detailed plans, reviewing with colleagues, seeking approval from the client, and conducting the process during off-hours to minimize the risk of downtime while delivering new features and security updates to mission-critical mainframe systems

Mainframe Programmer (Casual)
(Co-op)

May 2022 – August 2022
January 2021 – December 2021

Employment and Social Development Canada, Gatineau QC

- Produced regular reports for decision-makers in the organization using efficient COBOL programs executed with WFL on MCP mainframes to extract information from database files containing millions of records
- Supported a complex system of hundreds of interconnected COBOL programs by analyzing reported bugs and designing precise and robust fixes to maintain the existing functionality of mission-critical systems
- Updated documentation with clear and precise revisions following program updates to communicate changes to future programmers and foster a shared understanding among the development team
- Reverse-engineered a 7,000-line COBOL-76 program by deconstructing it into pseudocode and applying debugging tools to troubleshoot and document its functionalities, enabling an accurate rewrite in COBOL-84

Quality Assurance Analyst (Co-op)

September 2019 – December 2019

Global Affairs Canada, Gatineau QC

- Validated system specifications by reviewing and comparing documentation, identifying contradictions in software design to preempt issues later in development, helping the team deliver a quality product
- Modified and wrote new test cases in response to design changes, translating requirements and use cases into well-written, easy-to-follow, step-by-step guides to validate software ahead of release
- Performed black-box system testing by following requirement-linked test cases, documenting defects in bug tracking software to evaluate software quality and support developers in meeting design requirements
- Took initiative to compare and review related test cases with use case requirements in a spreadsheet to identify and report dozens of missing and superfluous tests, optimizing both testing efficiency and efficacy

PROJECTS

Astrosurfer Engine

February 2022 – April 2022

Personal Project, Ottawa ON

- Designed a real-time game engine implementing collision detection, physics simulation, and 3D graphics, developed from scratch with Python and OpenGL and using a property-centric architecture to optimize memory utilization and enable parallelization of game data updates on graphics hardware

Quest

February 2022 – April 2022

Object-Oriented Software Engineering Course Project, Carleton University, Ottawa ON

- Headed a team of three programmers and managed a shared Git repository, meeting weekly to solve problems collaboratively and coordinate programming tasks as well as review code to make deliverables on time in accordance with set deadlines
- Coordinated design and development of a digital version of a card game (*Quest*) in Java using object-oriented design principles and thoughtful application of design patterns to enhance code modularity and maintainability

EDUCATION

Bachelor of Computer Science (Honours), Computer Game Development Stream, Co-op option, Minor in History

September 2017 – December 2022

Carleton University, Ottawa ON

- Graduated With Distinction – CGPA 10.16/12.00 (A-)
-